Transpn Res. Vol. 3, pp. 397–408. Pergamon Press 1969. Printed in Great Britain

THE MINIMUM ROUTE PROBLEM FOR NETWORKS WITH TURN PENALTIES AND PROHIBITIONS

RONALD F. KIRBY
Planning Research Corporation, Washington, D.C., U.S.A.

and

RENFREY B. POTTS
Mathematics Department, University of Adelaide, South Australia

(*Received* 10 *March* 1969)

1. INTRODUCTION

THE TRANSPORTATION planning process has developed as an extremely complicated sequence of interrelated phases in which mathematical models of traffic assignment play an important role. These models assign traffic to minimum routes on road networks and it has been partly because of the availability of elegant algorithms for rapidly computing these routes that models of considerable sophistication have been widely used. An extensive bibliography of the literature on the minimum route problem has been prepared and recently revised by Murchland (1967).

It is important to distinguish between those transportation networks without turn penalties and prohibitions and those which take them into account. The former have been extensively used for main-road networks when the detailed traffic movements at individual intersections are unimportant. Many algorithms and variants of these for determining minimum routes on these networks have been proposed and the literature presents a very confusing picture which, however, has been greatly clarified by a recent article by Dreyfus (1967). Dreyfus gives an appraisal of the available algorithms and points out, for example, that some recently proposed algorithms are inferior to older ones, and that some published procedures are incorrect.

Dreyfus does not discuss networks with turn penalties and prohibitions and the present paper, which concentrates on these, is in a sense an extension of his article. It will be shown that the corresponding literature is equally confusing, that an important early publication has apparently been overlooked and that a recently published algorithm is incorrect. This paper concludes with a precise mathematical formulation of the problem and a discussion of some practical applications to transportation networks.

2. APPRAISAL OF PROPOSED PROCEDURES

In discussing the various procedures which have been suggested for determining minimum routes for networks with turn penalties and prohibitions, it could only add to the confusion to try to identify precisely the original authors. The following survey of the literature is not meant to be exhaustive but includes the main references known to the authors, listed in approximate chronological order.

(1) The first relevant publication of which the authors are aware is the short paper by Caldwell (1961). This paper seems to have been subsequently overlooked and has escaped

inclusion in the extensive bibliography of Murchland (1967). The simple argument contained in Caldwell's paper is essentially correct and he shows that turn penalties can be properly taken into account by constructing a pseudo-network in which nodes represent the original links and links represent 'hooks' or ordered pairs of links. Although turn prohibitions are not explicitly mentioned, they could be regarded as hooks with an infinite penalty. Caldwell also makes the following fundamental statement: "It isn't necessarily true that the best path from the origin to a node *i* through a node *j*, coincides, from the origin to *j*, with the best path from the origin to *j*." Algorithms which assume the contrary, which *is* valid for networks without turn penalties, are obviously incorrect. Caldwell illustrates his statement with a simple example which is shown in a modified form in Fig. 1(a). The numbers

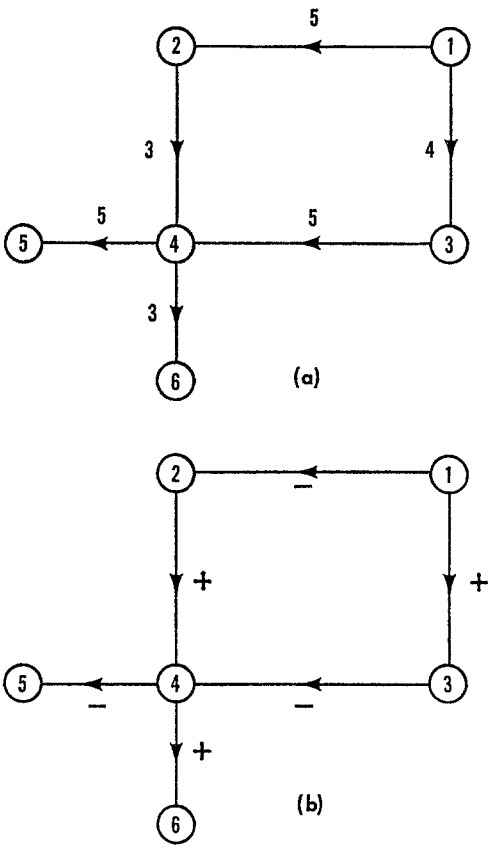


FIG. 1. (a) The cheapest route from node 1 to node 4 is 1, 2, 4 whereas the cheapest route from node 1 to 5 is 1, 3, 4, 5. Link costs are as indicated and all turns are penalized with a cost of 2 units. (b) The same network with links coded + and − to record changes of direction.

ascribed to the links are interpreted as link costs and each turn is assumed penalized with a cost of 2 units. The following route costs illustrate that 1, 2, 4 is the cheapest route from node 1 to node 4 whereas 1, 3, 4, 5 is the cheapest route from node 1 to node 5.

Perhaps one reason why Caldwell's paper has been neglected is because it does not formulate a practical algorithm which can be directly used for the computation of minimum routes for transportation networks as they are usually coded.

Route	Cost
1, 2, 4	5+2+3 = 10
1, 3, 4	4+2+5 = 11
1, 3, 4, 5	4+2+5+5 = 16
1, 2, 4, 5	5+2+3+2+5 = 17

(2) At about the same time that Caldwell's paper appeared, a suggestion was made by Martin *et al.* (1961) that turn penalties could be accounted for by the addition of a constant penalty for any 'change of direction' at an intersection. This suggestion has been incorporated in tree-building programs in transportation planning computer packages (Bureau of Public Roads, 1964; Control Data Corporation Data Centers Division, 1965) by labelling each north–south link with a 'plus' sign and each east–west link with a 'minus' sign as illustrated in Fig. 1(b). As the minimum route trees are built, a change in sign in passing from one link to another is used to indicate that a certain constant turn penalty should be added. Although this heuristic procedure tends to eliminate the building of 'illogical' staircase routes and is adequate for many networks, it has become well known by users of the programs that it is not strictly correct. For the example in Fig. 1(a), the algorithm would correctly obtain route 1, 2, 4 as the cheapest route from node 1 to node 4 but, having discarded the alternative route 1, 3, 4 would incorrectly obtain 1, 2, 4, 5 as the cheapest route from node 1 to node 5. The method has been modified, as suggested by Steel (1965), so that a variety of turn penalties can be included in specified intersection types, but the basic invalidity of the procedure is, of course, not overcome.

The tree-building programs in the same computer packages attempt to allow for turn prohibitions by separately listing them and excluding a route when a prohibition is indicated. In principle, therefore, these algorithms, like all those for networks without turn penalties and prohibitions, consider only those routes which are paths, strictly defined as sequences of distinct nodes with consecutive nodes joined by directed links. If, however, in Fig. 2(a), the left turn from link (6, 4) to link (4, 5) is prohibited, there is no path from node 6 to node 5 whereas the route *via* nodes 6, 4, 2, 1, 3, 4, 5 would be acceptable. (In practice, such routes are sometimes found by special features in the computer program. There are always networks for which these special features fail, however.) The importance of including such routes in transportation networks is exemplified by the use of signs in the heart of West London, England, which indicate to a motorist how he should negotiate a prohibited turn from an arterial road into a minor cross street. The signs, displayed on the approaches to the intersections, are a keep-to-the-left version of Fig. 2(b). The significance of Caldwell's pseudo-network in regard to this problem, although not discussed in his paper, is that the replacement of links by nodes implies that the class of routes admitted includes all those for which, on the original network, no *link* is traversed more than once. The route 6, 4, 2, 1, 3, 4, 5 is admissible because it is a sequence of distinct links (6, 4), (4, 2), (2, 1), (1, 3), (3, 4), (4, 5) even though it is not a path because node 4 is visited twice.

(3) One obvious way of correctly allowing for turn penalties and prohibitions is to represent intersections by sub-networks. For example, Wattleworth and Shuldiner (1963) and the ADECODE computer packages (SEMA, 1967) represent an intersection by 8 nodes and as many as 16 links [see Fig. 3(a)]. This representation gives a large network, without turn penalties and prohibitions, for which the standard minimum route algorithms are applicable. If all turning movements are allowed for at all intersections it is more efficient to use, as in the I.C.T. (1966) computer package, a representation of each intersection as 4 nodes and 12 links [see Fig. 3(b)]. This representation is equivalent to Caldwell's pseudo-network.

Although this added network structure removes the problem of turn penalties and prohibitions, it is of limited use in transportation applications because it introduces considerable difficulties in network coding and greatly increases the demands on computer storage. For these reasons, the method has not been widely accepted for incorporation in computer packages.

(4) A practical adaptation of Caldwell's pseudo-network or the I.C.T. network illustrated in Fig. 3(b) has been included in the IBM 360 Computer Programs as described

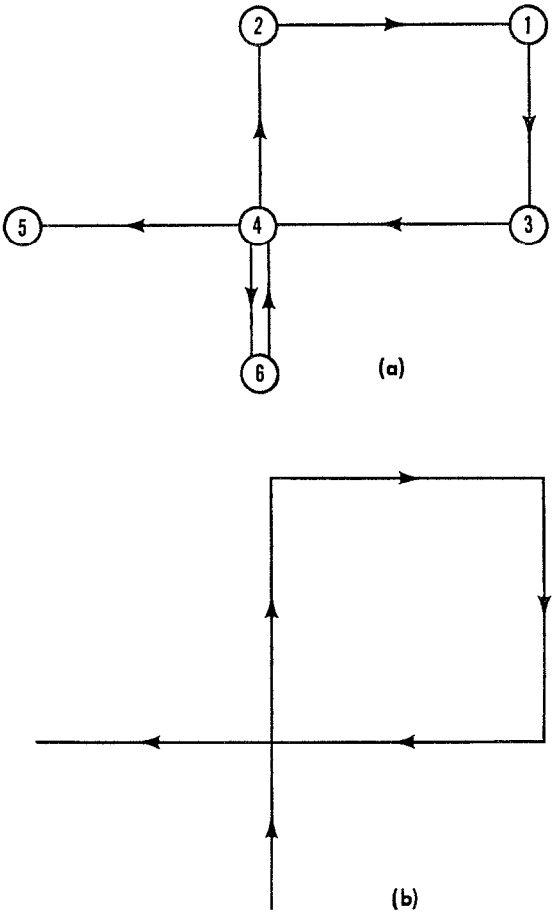


FIG. 2. (a) If the turn from link (6, 4) to link (4, 5) is prohibited, there is no path from node 6 to node 5 but there is an acceptable route 6, 4, 2, 1, 3, 4, 5 which visits node 4 twice. (b) Traffic sign indicating to a motorist how the prohibited turn should be negotiated.

by Skilton (1965). The tree-building program correctly allows for turn penalties and prohibitions by adding to each link a "leg" number 0, 1, 2 or 3 indicating the link to which the route is proceeding. As is usual practice, the number of links leaving a node is restricted to 4 but, in addition, the number of incoming links is likewise restricted so that the leg labelling is comprehensive. The resulting routes permit up to 4 visits to each node and the possible twisting and entwining of the routes have led to their description as "vines"

instead of the usual trees. Although the procedure described by Skilton is valid, he does not give a precise formulation of the algorithm used or include in his report any computational details.

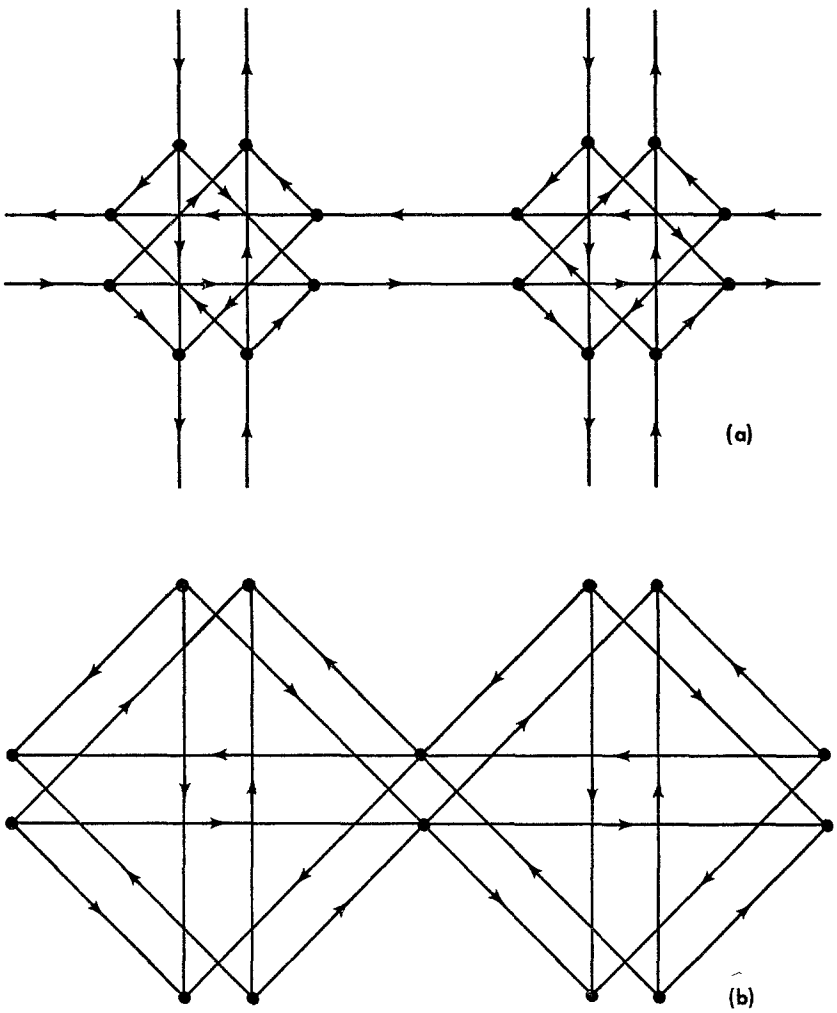


FIG. 3. (a) Network illustrating turning movements at two adjacent intersections. Note that U-turns are prohibited. Prohibition of other turns could be achieved by omitting corresponding links in the network. (b) A more efficient representation of two adjacent intersections. The midpoints of the roads are represented by nodes and the turning movements are included with travel on each half-road, thereby reducing the total number of nodes and links required.

(5) In a paper presented to the Third Australian Road Research Board Conference, one of the authors (Kirby 1966) described the computational aspects of an algorithm which is basically similar to that used in the IBM 360 programs and which correctly accounts for turn penalties and prohibitions. A fully documented version of the algorithm has been

lodged with the National Association of Australian State Road Authorities and has been used with success in metropolitan transportation studies.

(6) Although the state of the art at this stage would suggest that the problem of constructing valid practical route algorithms for networks with turn penalties and prohibitions was solved, the fact that confusion still exists is illustrated by a recent paper in this journal (Môri and Nishimura, 1967) which describes a procedure for determining the "shortest route under movement restrictions". The authors propose an algorithm based on the so-called matrix method, and as pointed out by Murchland (1968) in a brief note in this journal, their procedure is invalid. To add to the confusion, the same issue of the journal contains a paper by Butas (1968) in which the first of two suggested ways to take restricted turning movements into account is incorrect (page 260). In the hope of eliminating this confusion, we will in the next section consider two simple examples which illustrate the essential error in the procedure suggested by Môri and Nishimura.

This section may be summarized briefly by grouping the various proposed procedures in the following manner. The formulations of Caldwell, Skilton, I.C.T. and Kirby are mathematically equivalent and the most efficient methods proposed to date for correctly taking turn penalties and prohibitions into account. While I.C.T. and Caldwell describe the method in terms of pseudo-networks, Skilton and Kirby use the original network, simply finding shortest paths to each node via each possible entry link. This latter approach obviates the need to discuss a pseudo-network, which has always been an unattractive concept to transportation planners from the point of view of both coding and interpretation. For similar reasons the correct but involved method of Wattleworth and Shuldiner has not been widely used in transportation programs, and many unsuccessful attempts have been made to take account of turn penalties and prohibitions using the original road network with common shortest path algorithms. The methods of (2) and (6) above are typical of these latter approaches.

3. ILLUSTRATIVE EXAMPLES

As an illustration of the invalidity of some of the proposed procedures, two counter-examples will be described in detail to show why the method proposed by Môri and Nishimura fails. The matrix algorithm they use constructs a sequence of cost matrices $D^1, D^2, D^4, D^8, \ldots$ whose elements represent the costs of the cheapest routes between all pairs of nodes using not more than $1, 2, 4, 8, \ldots$ links. The elements of the matrix D^{2m} are calculated from the elements of the matrix D^m by the application of the formula

$$d_{ij}{}^{2m}(p,s) = \min_k [d_{ik}{}^m(p,r) + d_{kj}{}^m(q,s)] \tag{3.1}$$

where i, j represent nodes of the network, k an intermediate node, p, q "adjacent nodes of origin" and r, s "preceding nodes of destination". The elements of the initial matrix D^1 are the given link costs and the optimality is reached, for example, when $D^{2m} = D^m$. The authors eliminate a route with a prohibited turn by checking, for each intermediate node k, when the sequence of three nodes: (i) node preceding k, (ii) node k itself, (iii) node after k indicates a prohibition; if so, node k is eliminated in the application of equation (3.1).

The first simple example chosen to illustrate the error of this procedure is given in Fig. 4(a) which is an undirected network with link costs as indicated and with U-turns and the right turn from link $(1, 3)$ to link $(3, 4)$ prohibited. For simplicity we shall ignore the problem of determining the cheapest routes themselves and concentrate on their costs. The application of equation (3.1) leads to the matrices D^1, D^2 and D^4 as given in Tables 1, 2 and 3. An asterisk indicates no route or one of infinite cost. Because $D^4 = D^2$, these tables indicate optimality, and in particular that there is no route from node 1 to node 4. However, it is clear that there is a path $1, 2, 3, 4$.

As a second example, consider the undirected network illustrated in Fig. 4(b). The results of the application of the algorithm are given in Tables 4, 5 and 6. Again the equality of the last two tables indicates optimality and, in particular, that the cost of the cheapest route from node 1 to node 4 is 6 units via the link (1, 4). However, the cost of the path 1, 2, 3, 4 is only 5 units.

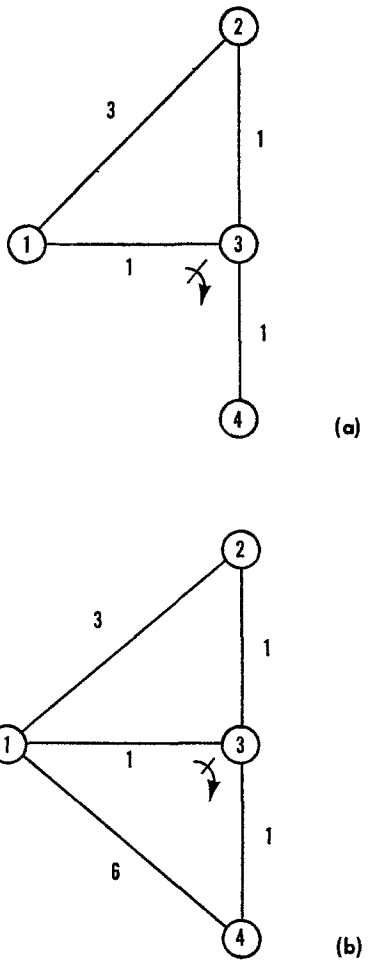


FIG. 4. (a) Network with indicated link costs and prohibited U-turns and a left turn. The costs of the cheapest routes between pairs of nodes is incorrectly calculated in Tables 1, 2 and 3. (b) The incorrect calculations for this network are given in Tables 4, 5 and 6.

It is not difficult to see, from these examples, why the algorithm, which may at first glance seem intuitively correct, is in fact invalid. In essence it does not heed the fundamental statement in Caldwell's paper quoted above. The link (1, 2) with cost 3 units is dropped from consideration because cheaper routes with no more than 2 links do not use it. Thus in Table 2, the (1, 2) matrix element corresponds to the route 1, 3, 2 of cost 2 units and the (2, 1) element to the route 2, 3, 2 of cost 2 units. The link (1, 2) is not included in any routes

TABLE 1. MATRIX D^1 FOR NETWORK FIG. 4(a)

O	D			
	1	2	3	3
1	0	3	1	*
2	3	0	1	*
3	1	1	0	1
4	*	*	1	0

TABLE 2. MATRIX D^2

O	D			
	1	2	3	4
1	0	2	1	*
2	2	0	1	2
3	1	1	0	1
4	2	2	1	0

TABLE 3. MATRIX D^4

O	D			
	1	2	3	4
1	0	2	1	*
2	2	0	1	2
3	1	1	0	1
4	2	2	1	0

TABLE 4. MATRIX D^1 FOR NETWORK FIG. 4(b)

O	D			
	1	2	3	4
1	0	3	1	6
2	3	0	1	*
3	1	1	0	1
4	6	*	1	0

TABLE 5. MATRIX D^2

O	D			
	1	2	3	4
1	0	2	1	6
2	2	0	1	2
3	1	1	0	1
4	2	2	1	0

TABLE 6. MATRIX D^4

O	D			
	1	2	3	4
1	0	2	1	6
2	2	0	1	2
3	1	1	0	1
4	2	2	1	0

with costs given by matrix D^2, and consequently it is never considered subsequently. Yet its final inclusion in the minimum routes is necessary.

The authors describe a labelling procedure for determining the minimum routes themselves by back-tracking, but it is clear from the above argument that their method will not work when turn prohibitions are present. Their method is even somewhat dubious when there are no prohibitions, because the authors do not explain how to cope with the ambiguity which arises in the definition of auxiliary nodes when the diagonal matrix elements are involved in their formulae.

4. MATHEMATICAL FORMULATION

It is apparent that one significant reason for the confusion in the literature on minimum route algorithms is the scarcity of precise mathematical formulations and justifications for the various procedures which have been proposed. Although an algorithm, when briefly described and illustrated perhaps by a single example, may seem intuitively correct, an exact formulation and proof of its validity are essential.

For networks with turn penalties and prohibitions, it would appear that the most useful and practical formulation is based on the definition of admissible routes as sequences of distinct links of the original network. Suppose that the directed network is denoted by $[N; L]$ with N the set of nodes $1, 2, 3, \ldots$ and L the set of directed links $l_1, l_2, l_3, \ldots$. Suppose further that the cost associated with a link l_i is denoted by $c(l_i)$ and the penalty associated with a turn from link l_i to link l_j by $p(l_i, l_j)$. For all l_i, l_j for which there is no permitted turn, $p(l_i, l_j) = \infty$. A finite value for $p(l_i, l_j)$ necessarily implies that $l_i \neq l_j$ and that l_i is directed to and l_j directed from the same node. An *admissible route* on the network can now be formally defined as a sequence $l_1, l_2, \ldots, l_k$ of links of L which are distinct, except possibly for l_1 and l_k, and are such that $p(l_i, l_{i+1}) < \infty$, $i = 1, 2, \ldots, k-1$. The special admissible route with $l_1 = l_k$ will be called a *loop*, including the degenerate case of an 'empty' route l_1, l_1 for which no links are traversed. Without loss of generality and with some advantage, the origin of the admissible route will be interpreted as being the beginning of the link l_1 (on the far side of the intersection represented by the nodes from which l_1 is directed) and likewise the destination will be taken to be the beginning of the link l_k. The cost $C(l_1, l_k)$ of an admissible route from l_1 to l_k is then defined by

$$C(l_1, l_k) = \sum_{i=1}^{k-1} c(l_i) + p(l_i, l_{i+1}) \tag{4.1}$$

For the degenerate case of an empty route, (4.1) is interpreted as defining zero cost.

The minimum route problem can now be formulated as follows: for a given origin link l_1, determine an admissible route from l_1 to any destination link $l_k \in L$ such that the cost of the route is not greater than the cost of any other admissible route from l_1 to l_k. The cheapest costs themselves will be denoted by $C^*(l_1, l_k)$.

The correct algorithms for solving this problem may be justified by appeal to the following

THEOREM. For given origin link l_1 and any destination link $l_k \in L$, the cheapest costs $C^*(l_1, l_k)$ are the unique solutions of the functional equations

$$C(l_1, l_k) = \min_{l_i \in B(l_k)} \{C(l_1, l_i) + c(l_i) + p(l_i, l_k)\}, \quad l_k \neq l_1 \tag{4.2}$$

$$C(l_1, l_1) = 0 \tag{4.3}$$

if and only if for every loop $l_1, l_2, \ldots, l_k = l_1$ the route cost is non-negative.

In (4.2), the notation $B(l_k)$ signifies the set of links before l_k defined by

$$B(l_k) = \{l_i \in L;\ p(l_i, l_k) < \infty\} \tag{4.4}$$

A rigorous proof of this theorem has been given by Kirby (1968), based on a proof by Bellman (1958) of a more specialized theorem, and a similar theorem, with links replaced by nodes, holds for networks without turn penalties and prohibitions. It is important to note that in the above formulation of the problem, the basic concept of Caldwell's pseudo-network is apparent in the definition of admissible routes and their costs.

For transportation networks, the link costs and turn penalties would in practice be non-negative so that the necessary and sufficient condition contained in the theorem would automatically be satisfied. The actual method used to solve the functional equations (4.2), (4.3) for these networks is not important and any of the algorithms developed for finding minimum routes for networks without turn penalties and prohibitions could be applied. The best to choose would depend on the size and the structure of the network under consideration.

5. APPLICATION TO TRANSPORTATION NETWORKS

The problem of including turn penalties and prohibitions is particularly important for transportation networks representing the central business district (CBD) of a metropolitan area. In planning a route within a CBD, a motorist has to reckon with one-way streets, turn prohibitions and traffic light phases governing movements at intersections. Figure 5

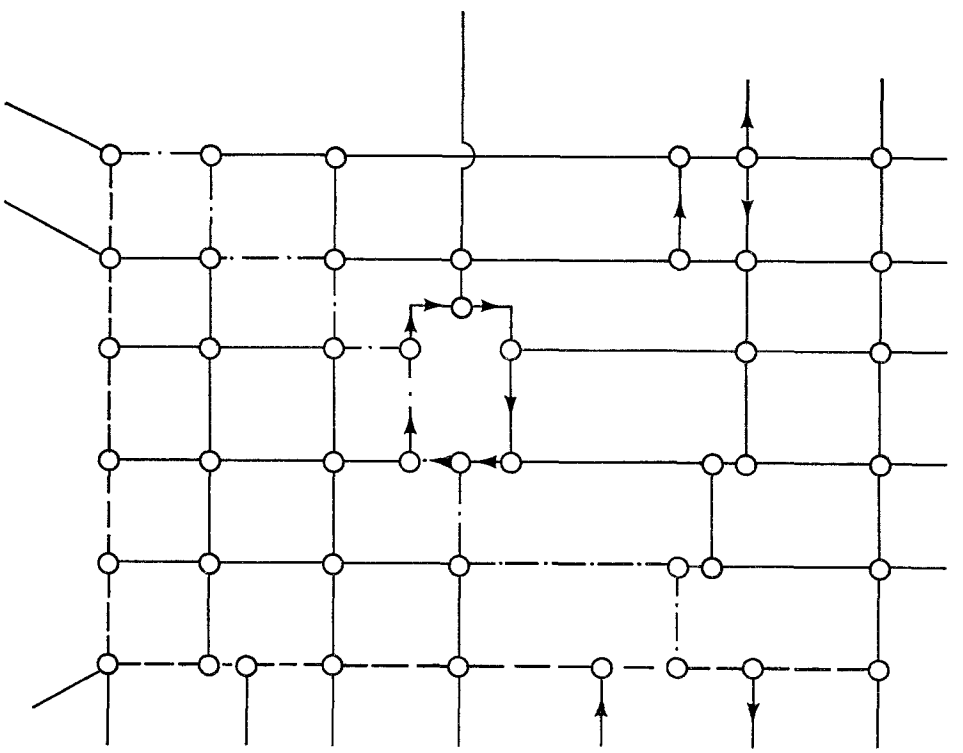


FIG. 5. Transportation network illustrating portion of the CBD of the City of Adelaide. The dashed lines correspond to the minimum route from the south-east corner to the north-west corner using an algorithm including turn penalties. The dash–dot lines correspond to the minimum route obtained ignoring turn penalties.

represents a portion of the CBD of the City of Adelaide and the minimum route from the node at the south-east corner to the node at the north-west corner has been calculated using the program written by Kirby (1966). The program contains an option which ignores turn penalties and prohibitions and with this option the illogical staircase route shown with dash–dot lines in Fig. 5 was obtained. The dashed line indicates the minimum route obtained with the option allowing for turn penalties.

It is interesting to note in the mathematical formulation of admissible routes given above that the origins and destinations are links rather than nodes. This concept has an important

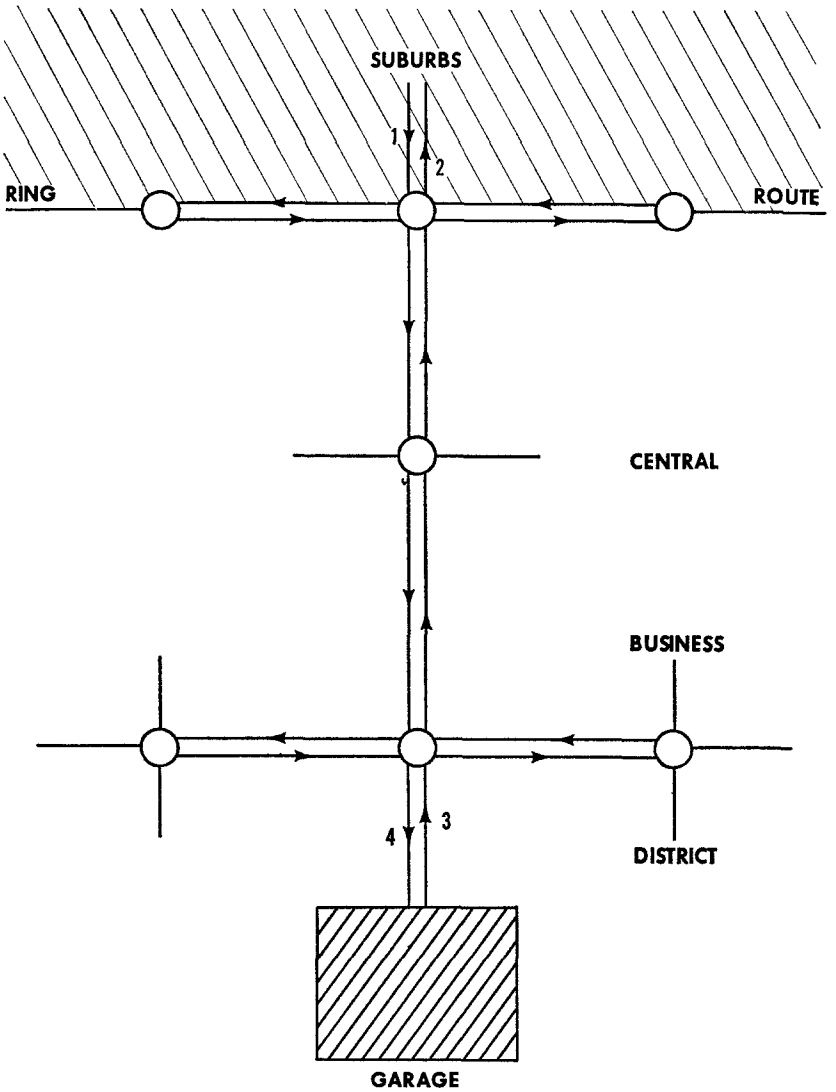


FIG. 6. Details of a portion of a CBD network showing origins and destinations of home-based work trips from the suburbs to a parking garage. Production: link 1, origin of a.m. trip; link 2, destination of p.m. trip. Attraction: link 3, origin of p.m. trip; link 4, destination of a.m. trip.

significance in considering routes on a CBD network for a motorist has to plan his route to end, for example, on a particular side of a city street or on the correct approach to a garage. The home-based work trips to city parking garages may be regarded as produced on links at a ring route, say, surrounding the CBD and attracted to the links at the locations of the garages as illustrated in Fig. 6.

In metropolitan transportation studies, planners have tended to concentrate on suburban regions because of the complexity of traffic movement in the down-town area. The valid procedures described in this paper for the determination of minimum routes on networks with turn penalties and prohibitions should form the basis of a traffic assignment model suitable for the analysis of the CBD. A complete description of the CBD network would include one-way streets, specifications for turn penalties and prohibitions and would be quite different for the a.m. peak, p.m. peak and mid-day periods. This detail can be readily incorporated in the minimum route algorithms and their proper use could be a significant aid to the planner and traffic engineer concerned with the detailed traffic movement on a city street network. An investigation along these lines is proposed for the City of Adelaide.

Acknowledgement—The authors have benefited from helpful correspondence with Mr. J. D. Murchland.

REFERENCES

BELLMAN R. E. (1958). On a routing problem. *Q. appl. Math.* **16**, 87–90.

BUREAU OF PUBLIC ROADS, U.S. Department of Commerce (1964). *Traffic Assignment Manual.*

BUTAS L. F. (1968). A directionally oriented shortest path algorithm. *Transpn Res.* **2**, 253–268.

CALDWELL T. (1961). On finding minimum routes in a network with turn penalties. *Communs Ass. comput. Mach.* **4**, 107–108.

CONTROL DATA CORPORATION DATA CENTERS DIVISION (1965). *Users' Manual, Transportation Planning System for the Control Data 3600 Computer.*

DREYFUS S. E. (1967). An appraisal of some shortest route algorithms. Rand Memorandum RM-5433-PR.

I.C.T. (1966). *1900 Series Scientific and Technical Software, Traffic Engineering.* Computer publication.

KIRBY R. F. (1966). A minimum path algorithm for a road network with turn penalties. *Proc. Third Aust. Rd Res. Bd Conf.*, Part 1, 434–442.

KIRBY R. F. (1968). Mathematical models for transportation planning. Ph.D. thesis, University of Adelaide.

MARTIN B. V., MEMMOTT F. W., III and BONE A. J. (1961). Principles and techniques of predicting future demand for urban area transportation. M.I.T. Report No. 3.

MÔRI M. and NISHIMURA T. (1967). Solution of the routing problem through a network by a matrix method with auxiliary nodes. *Transpn Res.* **1**, 165–180.

MURCHLAND J. D. (1967). Bibliography on the shortest route problem. London School of Economics, paper LSE-TNT-6.

MURCHLAND J. D. (1968). *Transpn Res.* **2**, 295–296.

SEMA (1967). *Le Modèle ADECODE.* Group Metra France, Paris.

SKILTON J. L. (1965). IBM urban planning programs. Presented at the Highway Engineering Exchange Program Meeting, Madison, Wisconsin.

STEEL M. A. (1965). Capacity restraint—a new technique. *Traff. Engng Control* **7**, 381–384.

WATTLEWORTH J. A. and SHULDINER P. W. (1963). Analytical methods in transportation; left-turn penalties in traffic assignment models. *J. Engng Mech. Div. Am. Soc. civ. Engrs* **89**, 97–126.